The Triple Barrier Labeling of Marco Lopez de Prado

Why Most Labels in Trading Are Wrong

LUCAS
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You want to train a model to predict good trades.
So what do you do? You create a label like:

> “If price goes up after 10 bars → label = 1, else 0.”

It sounds logical — but in reality, it’s flawed.

This kind of label ignores two fundamental things every real trade faces:

- **Risk** (what if the price drops before going up?)
- **Timing** (what if it goes up… but only after weeks?)

That’s why serious quants don’t use simple returns as labels.
They use something far more robust: the **Triple Barrier Method**.

In this newsletter, you’ll learn how it works, why it’s better, and how to implement it Python using the **quantreo** library, so your backtests stop lying and your models s learning.

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1. What Is the Triple Barrier Method?

The **Triple Barrier Method**, introduced by Marcos López de Prado, is a powerful w to label trading data.
Instead of checking whether the price goes up or down after *n* candles, it simulate **realistic trade conditions** using three key barriers:

🟢1. Upper Barrier → Take Profit

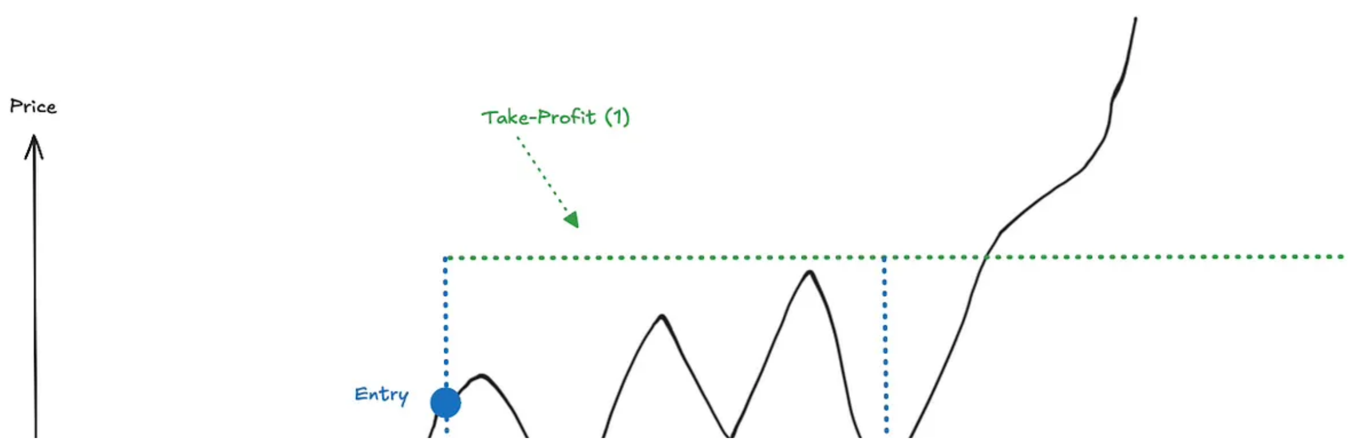
Set at a fixed percentage or return above the entry price.
If the price hits this level first, the trade is labeled as **+1** (profit).

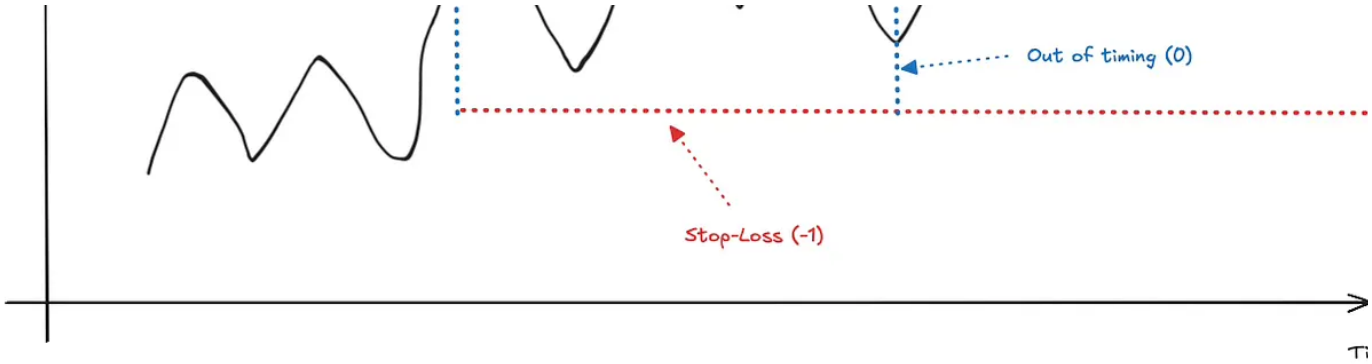
🔴2. Lower Barrier → Stop Loss

Placed below the entry price.
If this level is reached first, the label is **-1** (loss).

⏱️3. Vertical Barrier → Max Holding Time

If neither TP nor SL is hit within a certain number of periods, the trade **expires**, an the label is **0** (neutral).





📌**What makes it powerful?**

It doesn't just care about how much the price moved —
It asks **how the trade played out** under realistic conditions:

- Did it reach the profit fast enough?
- Was it stopped out?
- Did it just move sideways?

> The result is a label that reflects both the **direction**, the **risk**, and the **timing** of each opportunity.

And that makes it **ideal for machine learning**, especially in noisy markets.

2. How to Use the Triple Barrier Method in Practic

Now that you understand the concept, let's make it concrete.
You don't need to reinvent the wheel, the method is already implemented in the **quantreo** library.

Let's go through a simple example to label your data.

🧪Step 1: Load a Sample Dataset

```
from quantreo.datasets import load_generated_ohlcv_with_time

df = load_generated_ohlcv_with_time()
df = df.loc["2016"]
```

	open	high	low	close	volume	low_time	high_ti
time							
2016-01-04 00:00:00	104.944241	105.312073	104.929735	105.232289	576.805768	2016-01-04 03:21:00	2016-01-04 00:23
2016-01-04 04:00:00	105.233361	105.252139	105.047564	105.149357	485.696723	2016-01-04 04:15:00	2016-01-04 07:48
2016-01-04 08:00:00	105.159851	105.384745	105.141110	105.330306	403.969745	2016-01-04 09:07:00	2016-01-04 10:21
2016-01-04 12:00:00	105.330306	105.505799	104.894155	104.923404	1436.917324	2016-01-04 15:50:00	2016-01-04 12:01
2016-01-04 16:00:00	104.914147	105.023293	104.913252	105.014347	1177.672605	2016-01-04 18:22:00	2016-01-04 16:09
...	...	...	...	...	...	...	...
2016-12-30 04:00:00	103.632257	103.711884	103.495896	103.564574	563.932484	2016-12-30 05:34:00	2016-12-30 04:00
2016-12-30 08:00:00	103.564574	103.629321	103.555581	103.616731	697.707475	2016-12-30 10:00:00	2016-12-30 11:59
2016-12-30 12:00:00	103.615791	103.628165	103.496810	103.515847	1768.926665	2016-12-30 14:57:00	2016-12-30 12:00
2016-12-30 16:00:00	103.515847	103.692243	103.400370	103.422193	921.437054	2016-12-30 16:36:00	2016-12-30 17:43
2016-12-30 20:00:00	103.420285	103.429955	103.195921	103.226868	764.291608	2016-12-30 21:02:00	2016-12-30 20:00

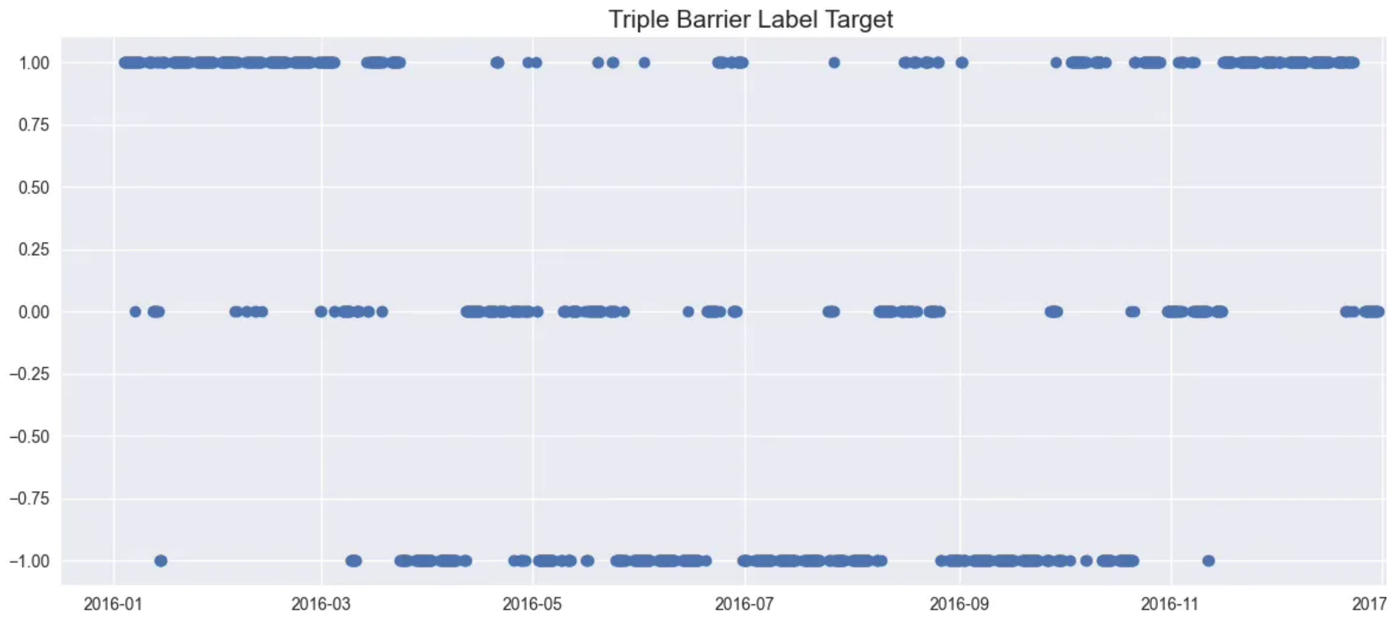
To use the Triple Barrier Method, you **must have** `low_time` **and** `high_time` **columns**, they tell when the price reached its lowest and highest points during the holding period.
If you're working with resampled data (like 1H or 4H candles), just extract 1-minute data first, then resample it; this way, you can easily track **when** the high or low occurred. If you use alternative bars, just use the ticks.

🧱Step 2: Apply the Triple Barrier Labeling

```
df["label"] = te.directional.triple_barrier_labeling(df, 500,
open_col="open", high_col="high", low_col="low",
high_time_col="high_time", low_time_col="low_time", tp=0.015, sl=-0.015
buy=True)
```

This function scans each row in the dataset, applies the three barriers (TP, SL, time) and creates a new column `label` with values:

- `1` → trade hit the **take-profit**
- `−1` → trade hit the **stop-loss**
- `0` → trade expired with neither hit



✅**That's it.**

Now your ML models won't just learn from oversimplified returns.
They'll learn from **real-world trade outcomes**, shaped by profit, loss, and time.

> The Triple Barrier Method gives your model a much stronger foundation to distinguish between **valid trades** and **noisy signals**.

3. Why It's a Perfect Fit for Meta-Labelling

The Triple Barrier Method is a **natural foundation** for building meta-labels.

In meta-labelling, your goal isn't to predict the market directly — it's to **decide whether to act** on a primary signal (long/short). But to do that, your model needs feedback on how those signals typically end.

That's exactly what the triple barrier provides.

It gives each entry signal a **structured label**:

- `+1` if the trade would have reached take-profit,
- `−1` if it would have hit the stop-loss,
- `0` if it expired before reaching either.

This clear outcome gives your meta-model a **concrete objective**: learn to **filter ou bad signals**, and only act on those with a higher chance of success.

In short, the Triple Barrier Method transforms raw directional signals into a reliable training set for smarter decision-making.

Most traders label their data in a simplistic way, and wonder why their models don' work.

The **Triple Barrier Method** changes the game.
It adds *realism* to your backtests by mimicking what would actually happen in a tra Take-profit, stop-loss, or timeout.

And if you're building advanced workflows like **meta-labelling**, it's not optional — it's the backbone that allows your meta-model to **learn from real trade outcomes** not noisy price shifts.

👉Start using it today with just a few lines of code in the **quantreo** library. Your model, and your future self, will thank you.

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